

# Mohamed Mdhaffar

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## Summary

Computer Vision & AI Systems candidate looking for an End-of-Year Internship. Combines expertise in PyTorch and autonomous AI agents with low-level CUDA optimization and real-time systems engineering to design low-latency ML architectures. 1st Place - GAIAFest Continental Finals, Accra (2025), representing Tunisia as National Champion.

## Education

**Higher School of Communications of Tunis - SUP'COM** **2024 – 2027**  
ICT engineering

**Preparatory institute for engineering studies of Sfax** **2022 – 2024**  
Mathematics and physics  
Ranked 74/1750 in the national entrance exam for engineering schools.

## Experience

**Dione Protocol and Carbon Insights** **July 2025 – August 2025**  
*Blockchain & AI Engineering Intern* *Tunis and Newark, US (Remote)*

- Built an AI-powered token on Dione Coin that automatically generates and publishes blockchain content on X using LLMs.

**CERT** **June 2025**  
*AI & IoT Web Development Intern* *Tunis, Tunisia*

- Developed an AI-powered carbon emission monitoring platform integrating a LoRaWAN IoT sensing node, Sentinel-5P satellite data, and a React/Django dashboard for real-time greenhouse gas monitoring and ML-based forecasting, enhanced with a RAG-based chatbot for natural-language querying of stored emission data.

**WAYCON** **July – August 2026**  
*Computer vision and ai agent Intern* *Tunis, Tunisia*

- Developed three AI systems: a real-time RTSP video analytics pipeline, a natural-language analytics agent for PostgreSQL insights and anomaly detection, and a vehicle analytics system for tracking, vehicle attributes, and Tunisian license plates.

## Projects

**Image Denoising and Deblurring software** **Research project**

- Built and optimized a zero-shot Deep Image Prior reconstruction pipeline using ADMM-TV/WTV, custom CUDA kernels, torch.compile, TorchInductor, and Triton, achieving up to  $9.5\times$  speedup on a A100 GPU.

**TSYP13** **2nd place**

- Built an AI-powered ADCS monitoring system for ESA's OPS-SAT, combining real-time telemetry, ANN/Random Forest anomaly detection, and STM32F429 edge deployment using STM32Cube.AI and Edge Impulse.

**Gemma 4 Kaggle Competition**

- Explored Gemma 4 and llama.cpp to design a low-latency sign language translation system for mobile devices using LiteRT, integrating an ST-GCN model with MediaPipe for real-time hand sign recognition

**Bio-Vector Orbit** **7th place**

- Built a biomedical RAG discovery engine combining semantic retrieval over papers, proteins, and molecules using Qdrant similarity search.

**Diabetic Retinopathy Detection**

- Developed an ML system for diabetic retinopathy detection with explainability, reliability checks, and secure medical image packaging.

## Skills and Languages

**AI & ML:** PyTorch, Computer Vision, LLMs, Transformers, CNNs, VAEs, GANs, LangChain, LangGraph, MCP.

**GPU & Edge AI:** CUDA C++, Triton, Numba, PyTorch Profiler, TorchInductor, FlashAttention, vLLM, STM32Cube.AI.

**Software & Backend:** Python, C/C++, JavaScript, Django, React, Qdrant, Docker, Git/GitHub, AWS EC2.

**Coursework:** Stanford CS229, CS230, CS231N, CME295, CS236, CS149, EE367/CS448I; GPU Mode CUDA.

**Research:** Exploring efficient LLM inference, GPU optimization, and neurosymbolic testing; studied Centaur, a fuzzer for deep learning libraries and reading state of the art research papers.

**Leadership:** IEEE GRSS HR Manager; Deep Learning and AI-based Supernova Detection workshop instructor.

**Languages:** Arabic (Native), French (Bilingual), English (Professional – TOEIC: 765/990).